

MS Word

Microsoft Word (MS Word) is a word processing software developed by Microsoft. It is widely used for creating, editing, formatting, and printing text documents such as letters, reports, resumes, and essays.

Key Components

Component	Description
Title Bar	Displays the name of the document.
Scroll Bars	Used to scroll through the document (vertical & horizontal).
Quick Access Toolbar	Offers quick access to frequently used commands like Save, Undo, and Redo.
Control Buttons	Minimize, Maximize, and Close buttons for the window.
Status Bar	Displays information like page number, word count, and language.
Rulers (Vertical & Horizontal)	Help in aligning text, images, and other elements.
Ribbon	The top panel that includes all tabs, groups, and commands.

Ribbon Structure:

Tabs → Groups → Commands

Shortcut Keys

Action	Shortcut
Save	Ctrl + S
Undo	Ctrl + Z
Redo	Ctrl + Y
New Document	Ctrl + N
Select All	Ctrl + A

Tabs in MS Word

There are 8 main tabs in the Ribbon:

- 1. Home (Alt + H)**
- 2. Insert (Alt + N)**
- 3. Design (Alt + G)**
- 4. Layout (Alt + P)**
- 5. References (Alt + S)**
- 6. Mailings (Alt + M)**
- 7. Review (Alt + R)**
- 8. View (Alt + W)**

Home Tab

Clipboard

- **Cut (Ctrl + X):** Removes selected text and stores it on the clipboard.
- **Copy (Ctrl + C):** Copies selected text to the clipboard.
- **Paste (Ctrl + V):** Inserts the last copied or cut content.
- **Paste Special (Ctrl + Alt + V):** Pastes content in a specific format (e.g., plain text, image).
- **Format Painter:** Copies formatting from one section and applies it to another.
 - **Copy Formatting:** Ctrl + Shift + C
 - **Paste Formatting:** Ctrl + Shift + V

Note: Clipboard can store up to **24 items**.

Font

- **Change Font (Ctrl + Shift + F):** Change text style (e.g., Arial, Times New Roman).
- **Change Font Size (Ctrl + Shift + P)**
- **Bold (Ctrl + B), Italic (Ctrl + I), Underline (Ctrl + U)**
- **Grow Font (Ctrl + Shift + >), Shrink Font (Ctrl + Shift + <)**
- **Grow Font by 1 point (Ctrl + Shift +]), Shrink Font by 1 point (Ctrl + Shift + [)**
- **Font Dialog Box (Ctrl + D):** Open advanced font settings.
- **Strikethrough, Subscript (Ctrl + =), Superscript (Ctrl + Shift + +)**
- **Font Color & Highlighter**
- **Text Effects** (e.g., Shadow, Glow)

Color Types:

- **Theme Colors:** Changeable with document themes.
 - **Standard Colors:** Fixed and unchangeable.
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Paragraph

- **Bullets, Numbering, Multilevel List**
- **Indent (Increase/Decrease)**
- **Text Alignment:**
 - **Left (Ctrl + L)**

- Center (Ctrl + E)
 - Right (Ctrl + R)
 - Justify (Ctrl + J)
 - **Line & Paragraph Spacing**
 - **Borders**
 - **Show/Hide Formatting Marks (Ctrl + *)*
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Styles

- **Apply Styles (Ctrl + Shift + S)**
 - **Create Style / Save Selection as Quick Style**
 - **Clear Formatting**
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Editing

- **Find (Ctrl + F):** Open Navigation Pane with Headings, Pages, and Search Results.
- **Replace (Ctrl + H):** Replace text or characters in a document.
- **Select Options:**
 - Select All
 - Select Objects
 - Select Similar Formatting

Special Tip – Remove Paragraph Spaces:

- **Find:** ^p
- **Replace:** (Leave blank or use space)

Insert Tab

Pages

- **Cover Page:** Adds a pre-designed title page.
 - **Blank Page:** Inserts a new blank page.
 - **Page Break (Ctrl + Enter):** Moves content after the cursor to a new page.
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Tables

Notes:

→ Maximum: 10 rows and 8 columns

Table Design

Table Style Options

- Header Rows
- First Columns
- Total Row
- Last Column
- Banded Rows
- Banded Columns

Table Styles

- Plain Tables
- Grid Tables
- List Tables

Borders

- Border Styles
 - Line Style
 - Line Weight
 - Pen Color
 - Borders
 - Border Painter
-

Layout

Table

- Select

i) Cell, Columns, Row & Table

- View Gridlines
- Properties

i) Table, Column, Row & Cell

Draw

- Draw Table
- Eraser

Rows & Columns

- Delete
- Insert (Above, Below, Left, Right)

Merge

- Merge Cells
- Split Cells
- Split Table

Cell Size

- Autofit
- i) Content, Window, Column Width
- Height
 - Width
 - Distribute Rows
 - Distribute Columns

Alignment

- Top, Center, Bottom
- i) Left
- ii) Center
- iii) Right
- Text Directions
 - Cell Margins

Data

- Sort
- Repeat Header Rows
- Convert to Text
- Formula

Illustrations

- **Pictures:** Insert images from your device.
- **Shapes:** Add predefined shapes like arrows, circles, etc.
- **SmartArt:** Visual representations like flowcharts or processes.

- **Charts:** Graphs (bar, line, pie, etc.).
 - **Screenshot:** Insert a screenshot of open windows (Fn + PrtScn).
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Links

- **Hyperlink (Ctrl + K):** Insert a clickable link.
 - Link to: Web, another file, email, location in document
 - **Bookmark:** Mark a specific location to quickly return to.
 - **Cross-reference:** Link to Headings, Footnotes, Figures, Tables, etc.
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Header & Footer

- **Header/Footer:** Repeating text at top/bottom of each page.
 - **Page Number:** Insert page numbers with different styles/positions.
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Text

- **Text Box:** Insert movable text blocks.
 - **Quick Parts:** Reuse content blocks (e.g., author name, document title).
 - **WordArt:** Stylized decorative text.
 - **Drop Cap:** Large capital letter at paragraph start.
 - **Signature Line:** Add electronic signature space.
 - **Date & Time**
 - **Object:** Insert non-text items (Excel sheets, PDFs, etc.).
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Symbols

- **Equation (Alt + =):** Insert pre-made or custom math equations.
 - **Symbol:** Insert special characters like ©, ®, etc.
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Design Tab

Document Formatting

- **Themes:** Change the overall design (colors, fonts, effects).
- **Colors & Fonts:** Pick new color schemes or font sets.
- **Paragraph Spacing:** Apply spacing between lines/paragraphs.
- **Effects:** Apply 3D, shadows, and reflections to shapes.

Page Background

- **Watermark:** Add "Confidential", "Draft", or custom text as background.
 - **Page Color:** Change document background color.
 - **Page Borders:** Decorate page edges.
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Layout Tab

Page Setup

- **Margins:** Adjust space around the text.
 - **Orientation:** Switch between Portrait or Landscape layout.
 - **Size:** Select standard sizes (A4, Letter, etc.).
 - **Columns:** Split text into 1–13 columns.
 - **Breaks:** Add section, page, or column breaks.
 - **Line Numbers:** Add numbering to each line.
 - **Hyphenation:** Automatically break words across lines.
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Paragraph

- **Indent:** Shift text away from margins.
 - i) Left
 - ii) Right
 - **Spacing:** Adjust space before/after paragraphs.
 - i) Before
 - ii) After
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Arrange

- **Position:** Move object to specific position.
 - **Wrap Text:** Set how text wraps around images/objects.
 - **Bring Forward / Send Backward:** Layer objects.
 - **Selection Pane:** List of all your objects
 - **Align:** Line up objects vertically/horizontally.
 - **Group/Ungroup:** Combine or separate multiple objects.
 - **Rotate:** Rotate images, text boxes, or shapes.
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References Tab

Table of Contents

- **Insert Table of Contents:** Automatically generated from headings.
 - **Add Text:** Mark headings to appear in the TOC.
 - **Update Table:** Refresh TOC after changes.
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Footnotes

- **Insert Footnote (Alt + Ctrl + F):** Notes at the bottom of the page.
 - **Insert Endnote (Alt + Ctrl + D):** Notes at the end of the document.
 - **Next Footnote / Show Notes**
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Citations & Bibliography

- **Insert Citation:** Add book/article references.
 - **Manage Sources:** Edit or reuse references.
 - **Bibliography:** Auto-generate based on citations.
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Captions

- **Insert Caption:** Label images/tables (e.g., "Figure 1").
 - **Insert Table of Figures**
 - **Update Table**
 - **Cross-reference:** Link to labeled items.
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Index

- **Mark Entry (Alt + Shift + X):** Highlight important terms.
 - **Insert Index:** Create a list of marked terms.
 - **Update Index**
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Table of Authorities

- **Mark Citation (Alt + Shift + I):** Legal citations.
- **Insert Table of Authorities**

Mail Merge

Steps of Mail Merge:

1. **Select Document Type**
 - Letter, Email, Labels, Envelopes, Directory
 2. **Start Document**
 - Use current, template, or existing document
 3. **Select Recipients**
 - Type a list, use existing file, or Outlook contacts
 4. **Write Letter**
 - Insert personalized fields: Name, Address Block, Greeting Line
 5. **Preview Letter**
 - View for each recipient, edit list if needed
 6. **Complete Merge**
 - Print or create separate documents
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Review Tab

Proofing

- **Spelling & Grammar (F7)**
 - **Thesaurus (Shift + F7)**
 - **Word Count**
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Comments

- Add, Delete, Navigate (Next/Previous), Show Comments
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Tracking

- **Track Changes (Ctrl + Shift + E)**
 - View changes as Simple/All Markup
 - Use Reviewing Pane for a summary
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Changes

- Accept, Reject changes one by one or all at once
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Restrict Editing

1. **Formatting Restrictions:** Limit formatting to a selection style
2. **Editing Restrictions:** Allow only this type of editing in the editing in the document.
 - iii) No Changes (Read Only)
 - iv) Comments
 - v) Filling in the form
 - vi) Track Changes
- **Exceptions (optional):** Everyone
3. **Start Enforcement**

View Tab

Document Views

- Read Mode, Print Layout, Web Layout, Outline, Draft
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Show Group

- Toggle **Ruler**, **Gridlines**, **Navigation Pane**
-

Zoom

- **Maximum Zoom:** 500%
 - **Medium Zoom:** 100%
 - **Minimum Zoom:** 10%
 - **One Page, Multiple Pages, Page Width**
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Window

- **New Window**
 - **Arrange All**
 - **Split (Ctrl + Alt + S)**
 - i) Side by Side View
 - ii) Synchronous Scrolling
 - **Switch Windows**
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Macros

- **View Macros (Alt + F8)**
- **Record Macros** for automating repetitive tasks
- **Pause Recording**

MS Excel

Microsoft Excel is a spreadsheet program used to store, organize, and analyze data. It's ideal for performing calculations, creating charts, managing data, and applying complex formulas.

Key Concepts

Term	Description
Workbook	A file that contains multiple worksheets.
Worksheet	A single spreadsheet made up of rows and columns.
Rows	Horizontal (numbered 1, 2, 3...) – Max: 1,048,576
Columns	Vertical (labeled A, B, C...) – Max: 16,384

Tabs in MS Excel

1. **Home**
 2. **Insert**
 3. **Page Layout**
 4. **Formulas**
 5. **Data**
 6. **Review**
 7. **View**
-

Addition

- **Method 1:**
=C1 + C2 + C3 → Add specific cells
- **Method 2:**
=SUM(C1:C4) → Add a range
- **Method 3:**
Drag to select cells and use =SUM(...)
- **Method 4:**
Select cell → Press Alt + = → AutoSum

Division

=IF(percentage>=60, "First", IF(percentage>=45, "Second", IF(percentage>=33, "Third", "Fail")))

Percentage

- =Obtained Marks / Total Marks * 100
 - =Obtained Marks / Number of Subjects
-

Count Functions

Function	Description
=COUNT(range)	Count numeric values only
=COUNTA(range)	Count all non-empty cells
=COUNTBLANK(range)	Count empty cells
=COUNTA - COUNT	Count text/alphabet cells

COUNTIF

- Counts the number of cells within a range that meets the given condition.

=countif(range,criteria)

COUNTIFS

- Counts the number of cells specified by a given set of conditions or criteria.

=countifs(criteria_range1,criteria1,criteria_range2,criteria2)

SUMIF

- Adds the cells specified by a given condition or criteria.

=sumif(range,criteria,[sum_range])

SUMIFS

- Adds the cells specified by a given set of conditions or criteria.

=sumifs(sum_range,criteria_range1,criteria1,criteria_range2,criteria2...)

GST Calculation

Tax Type	Usage	Example Rate
CGST	Central Govt. (Purchase)	2.5% – 14%
SGST	State Govt. (Purchase)	2.5% – 14%
IGST	Integrated (Sell)	12% – 28%

Purchase

= (Final Amount - Discount) + (CGST + SGST)

Sells

= (Total Amount - Discount) + IGST

Date/Time

Function	Description
=NOW()	Returns current date & time
=TODAY()	Returns current date
=DATEDIF(start_date, TODAY(), "y")	Calculates age in years
=DAY(date) / =MONTH(date) / =YEAR(date)	Extracts day, month, or year

Text

Function	Description	Example
=CONCATENATE(C6, " ", D6)	Join names	John Doe
=LEFT(A2, 4)	First 4 letters	John
=RIGHT(A2, 3)	Last 3 letters	Doe
= "Miss. " & B6	Add prefix	Miss. Jane
=CONCATENATE(E6, " S/O ", F6)	Combine fields	Miss. Jane S/O Mrs. Anne

Example:

=IF(C6="Male", "Mr. "&B6, IF(C6="Female", "Miss. "&B6))

Special Copying Tip

To copy values **without formulas**:

- Copy → Right-click → **Paste Special** → Choose **Values**
-

Math Functions

Function	Description
=ABS(A1)	Converts negative to positive
=ROUND(A1, 0)	Rounds off number to nearest integer

Remove Duplicates

- Select range → Data Tab → Click **Remove Duplicates**
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Advanced Filter

Used to copy and paste specific data based on criteria:

- List Range: Data source
 - Criteria Range: Filter condition
 - Copy To: Destination range
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Name Manager

- Assign a name to a cell or range (e.g., A1 = "Price")
 - Use in formula like =Price * Quantity
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Trace Precedents/Dependents

- **Precedents:** Show which cells affect a formula.
 - **Dependents:** Show where a cell is used in formulas.
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Evaluate Formula

- Step-by-step view of how Excel calculates a formula.
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Show Formula

- Shows formulas in cells instead of results.
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Text to Columns

Split text in a cell into separate columns:

- Go to Data → Text to Columns
 - Choose:
 - **Delimited:** Separate by comma, space, etc.
 - **Fixed Width:** Divide at set character length
-

What-If Analysis

Tool	Purpose
Data Table	Analyze multiple outputs for one formula
Goal Seek	Find required input for a desired result

Tool	Purpose
Scenario Manager	Compare different business cases

Goal Seek Example:

- Set total to 5000 by changing the rate.

Consolidate

- Combine values from multiple sheets or ranges.

Group/Ungroup

- Collapse or expand sections of rows/columns.

Subtotal

- Add automatic subtotals for grouped data.

Data Validation

Restricts invalid entries in cells.

Option	Description
Settings	Set min/max values, source list
Input Message	Pop-up when cell is selected
Error Alert	Shows a message on invalid input

Pivot Table

Summarizes large data sets.

Steps:

1. Select data

2. Insert Pivot Table
 3. Use fields to drag:
 - **Rows, Columns, Values, Filters**
-

Slicer

- Interactive filter for Pivot Table.
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Chart Types

- **Column/Bar, Line/Area, Pie/Doughnut, Radar, Combo**, etc.
-

Sparkline

Tiny charts inside a cell (Line, Column, Win/Loss).

Conditional Formatting

Highlights cells using rules:

- **Highlight Rules:** Greater than, equal to, text contains, etc.
 - **Top/Bottom Rules:** Top 10, bottom 10, above average, etc.
-

VLOOKUP (Vertical Lookup)

Searches for a value **vertically** in the first column and returns a value from the same row.

Syntax:

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

Example:

	A	B
	ID	Name
	101	John
	102	Sarah

To get name of ID 102:

=VLOOKUP(102, A2:B3, 2, FALSE) → Sarah

HLOOKUP (Horizontal Lookup)

Searches for a value **horizontally** in the first row and returns a value from the same column.

Syntax:

=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])

Example:

	A	B	C
	ID	101	102
	Name	John	Sarah

To get the name for ID 102:

=HLOOKUP(102, A1:C2, 2, FALSE) → Sarah
